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Effect of a dynamic seat pan design on spine biomechanics, calf circumference and perceived pain during prolonged sitting

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ABSTRACT

This study investigates the effects of a dynamic seat pan design on sitting biomechanics, perceived pain and seat movement compared to a control. Thirty male participants were recruited for two experimental sessions consisting of a 2-h sitting exposure (standardized typing task). Spine angles, back muscle activity, perceived pain and calf circumference were measured pre and post exposure. Sitting in the dynamic condition resulted in lower pain ratings ($p = 0.031$), decreased calf circumference ($p < 0.001$), lower average seat pressure ($p < 0.001$), and greater seat contact area ($p = 0.003$) compared to the control. Spine angles and low back EMG for all 6 muscles showed no significant differences between chair conditions. These results suggest this dynamic seat pan design is effective at decreasing several negative components associated with sitting for the occupant. Future work should examine the longer-term effects of dynamic office chair features in the field setting with a more generalizable population.

1. Introduction

Workers in the developed world spend one-third of the workday seated (Copeland et al., 2015). The resulting decreased metabolic demand of these postures, when not offset by physical activity, increases the odds of disease and early mortality (Ekelund et al., 2016). Reducing sitting time and taking activity breaks have been suggested to offset these adverse effects, however, this is not always feasible for some occupations (e.g., air traffic control personnel, call centre operators etc.) or acted upon by those free to move but engrossed in their work. Dynamic chair designs that facilitate movement while seated may be an ideal alternative for those that do not leave their chair frequently. Several previous authors have investigated the ability of uniaxial dynamic chairs to affect a range of variables such as spinal posture, muscle activation, and overall movement (Gregory et al., 2006; Kingma and van Dieën, 2009; O'Sullivan et al., 2013; O'Sullivan et al., 2012; van Dieen et al., 2001) and found that designs increase movement (Ellegast et al., 2012) and offset spine compression (van Dieen et al., 2001; van Deursen et al., 2000). Multiaxial dynamic chairs have also been shown to raise energy expenditure (Koepp et al., 2016) and facilitate movement, reduce calf swelling and improve cognitive attention (Triglav et al., 2019) in a laboratory setting. Thus, it appears that chair designs that allow for movement while sitting have the potential to mitigate the adverse metabolic, cardiovascular and musculoskeletal effects of prolonged seated exposures. At this time, it is unclear if these improvements are due to increased whole body movement and metabolic demand or other aspects such as reduced flexion and/or increased joint motion throughout the spine and hips.

There are several potential issues with the general lack of movement in sitting. Static sitting, where flexion at the spine and hips is held for long periods of time, has been associated with an escalation in hydrostatic pressure (Pottier et al., 1969), reduced blood flow (Restaino et al., 2015, 2016) and decreased shear stress on the vasculature (Restaino et al., 2016) which in turn has been suggested to increase the susceptibility of the leg vasculature to atherosclerosis and other serious cardiovascular problems (Restaino et al., 2016). While the mechanism of these associations is poorly understood, it is likely that lack of alternating muscle contractions of lower limb muscles together with compression at the hip and knee play a role. Indeed, sitting time itself is considered a risk factor for cardiovascular disease (Chomistek et al., 2013; Dunstan et al., 2012) and other related adverse health effects including increased risk of mortality (Dunstan et al., 2012; Katzmarzyk et al., 2009) type 2 diabetes (Dunstan et al., 2012), and certain cancers (Katzmarzyk et al., 2009). It is hypothesized that reducing static postures, such as by allowing for movement during sitting, minimizes endothelial dysfunction of the leg vasculature thus improving

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hemodynamics and reducing cardiovascular disease risk (Restaino et al., 2015; Duvivier et al., 2018); a potential avenue for future prospective cohort studies to investigate.

Another issue with prolonged static sitting is the potential to induce perceived musculoskeletal pain in some individuals. Although the association between clinical episodes of low back pain (LBP) and prolonged sitting is unclear (O'Sullivan et al., 2012; Aota et al., 2007; Danquah et al., 2017; Gupta et al., 2015; Chen et al., 2009; Hartvigsen et al., 2000), cross-sectional studies have consistently shown that sitting exposures induce clinically relevant levels of transient LBP in a proportion of adults regardless of clinical history in both the laboratory and field settings (De Carvalho et al., 2020; Greene et al., 2019). Reducing the static nature of sitting, with small postural movements of the spine such as fidgets and shifts, appears to play a role in modulating perceived pain (De Carvalho et al., 2020; Greene et al., 2019; Vergara and Page, 2002; Dunk and Callaghan, 2010). Therefore, it appears logical that a chair design that permits increased movement of spine joints (flexion/extension and lateral bend) during sitting may be helpful to prevent or minimize LBP.

Therefore, the purpose of this study was to explore the effect of a dynamic seat pan design, that permits 360 degrees of movement about a pivot point in the transverse plane, on lower limb swelling, spine posture, and perceived transient LBP. Our first objective was to compare calf circumference, as an indirect measure of lower limb venous pooling, between the dynamic design and a standard office chair. We hypothesized that the freedom for movement and seat pan orientation enabled by the pivot mechanism of the dynamic design would mitigate lower limb swelling due to increased movement of the hip and legs. Specifically, we expected that occupants would be using the muscles of their lower limb to change their postures during sitting and that this would translate to reduced venous pooling during the sitting exposure in the dynamic condition compared to control. The second objective was to compare the effect of the dynamic seat pan feature on biomechanical factors related to movement and posture. Here, we hypothesized that the ability to tilt the seat pan in any position about the pivot point would both result in the occupant self-selecting a seat pan orientation that would reduce spine flexion as well as increase postural movements of the spine and hips (active sitting) throughout 2 h of sitting compared to a standard chair. Lastly, we hypothesized that the combined effects of increased hip/spine movement and decreased spinal flexion would mitigate the development of LBP. This study will extend our understanding of the potential benefits of a dynamic seat pan design on cardiovascular, biomechanical, and perceived pain measures at the basic science level.

2. Methods

2.1. Participants

The required sample size to achieve a power of 0.80 with $\alpha = 0.05$, was calculated to be 28 participants using data from Triglav et al. (2019). Based on this calculation, 30 participants were recruited from the university population to account for dropouts. To reduce heterogeneity in this sample, only males were included for this study. Inclusion criteria were adult males between 18 and 69 years of age. Exclusion criteria included: a previous history of LBP linked to tumor, infection, fracture, or inflammatory arthropathy, and/or previous surgeries of the spine; inability to sit for 2 h at a time; an episode of LBP resulting in a lost day of work or school, in the past 6 months; or inability to attend both laboratory sessions. This study received ethics approval from the Health Research Ethics Board of Newfoundland and Labrador and all participants gave informed consent prior to starting the study.

2.2. Chairs

2.2.1. Dynamic chair

The dynamic condition (Fig. 1 - CoreChair, Core Chair Inc., Aurora, ON, Canada - https://corechair.com/our-core-design/) was a modified 5-caster office chair designed to facilitate active movement through a seat pan with a pivot mechanism directly under the seat that allows for movement in all directions (to a maximum tilt of 14°) about the transverse plane. The movement resistance was adjustable and, for standardization, set to level 2 of 5 for every participant. The seat pan was further sculpted and covered in 2″ thick foam padding to reduce seat pressure and provide stability. The backrest was low (9”) and adjustable in depth to stabilize the pelvis.

2.2.2. Control chair

The control condition used a standard 5-caster office chair with a tall back rest and built-in lumbar support (geoCentric Mid-Back Multi-tilt, ergoCentric Seating Systems, Mississauga, ON, Canada). The armrests were removed to create a condition more like the dynamic condition.

2.3. Instrumentation, processing & analysis

2.3.1. Calf circumference

Calf circumference has been used as an indirect measure of venous pooling given its relation to volume changes in the lower limb (Chester et al., 2002). To do this, the experimenter measured and marked a point 10 cm distal to the lowest aspect of the patella on the participant's dominant leg with a pen. This point was used to estimate the widest aspect of the calf, consistent with previous studies (Triglav et al., 2019; Restaino et al., 2015; Chester et al., 2002). Circumference was taken to the nearest mm using a clinical measuring tape by same researcher. Three measures were taken at each time point. If one of the three measures differed by 0.5 cm compared to the other two measures, a fourth was taken and the outlier was dropped. The average circumference of the three measures was calculated for each time point. Calf-circumference change over the exposure was calculated by subtracting the pre-sitting measure from the post-sitting measure.

2.3.2. Tri-axial accelerometers to measure spine angles and movements of the spine and seat pan

Two tri-axial accelerometers (ADXL335, Analog Devices, Norwood, MA, USA) were taped to the skin of the participant overlying the first lumbar and second sacral spinous processes in the +y down, +z anterior orientation to measure spine angles continuously through the experiment. Spinous process location was identified by a trained researcher using standard surface palpation techniques (Snider et al., 2011) and confirmed by a clinician (DDC) where necessary. A third tri-axial accelerometer was fixed to the respective chair's rigid arm of the backrest at a point closest to the seat-pan using industrial grade tape. Accelerometers were removed after each collection, but placement was standardized with a permanent marker outlining the location on the chair. Accelerometer data were low-pass filtered at 500 Hz, and A/D converted using a 16-bit board at a sampling frequency of 1500 Hz (DTS, Noraxon, Phoenix, AZ, USA).

Accelerometer data were processed using custom software (Matlab version 2017b, The Mathworks Inc., Natick, Massachusetts, USA) which included calibrating with respect to gravity, converting voltages to

Fig. 1. Schematic of the tested multiaxial dynamic chair (CoreChair, CoreChair Inc., Aurora, ON, Canada). Pivot point is located at the base of the seat pan.

accelerations, calculating absolute angles, smoothing the data using a dual-pass 2nd order Butterworth filter with a cut-off frequency of 1Hz, and determining the relative lumbar, pelvic, and seat pan angles from the absolute angles. Normalized versions of the lumbar angles were then calculated using the posture calibration trials to express spine angles as a percentage of maximum flexion range of motion (% ROM: 0% = maximum extension; 100% = maximum flexion). The following variables were used for the analysis: normalized lumbar spine angle (% ROM), pelvic angle (° with regards to standing), sagittal (front to back) and coronal (side to side) chair angle (change in ° with regards to empty chair), standard deviations of the sagittal and coronal chair angle (change in ° with regards to empty chair), and the range of the sagittal and coronal of chair angle respectively (°). Each variable was calculated over eight 15-min blocks throughout the trial.

To analyze the frequency of spine movements over each prolonged sitting trial, the number of fidgets (small change in posture immediately followed by a return to the same position) were calculated. The number of fidgets was calculated from the time-varying signal using established methods from the literature (Gallagher and Callaghan, 2015). Specifically, fidgets were defined as a movement of ± 3 SD or larger for a duration of 5 s or less. The number (frequency) and magnitude (size in degrees) of fidgets during the tenth minute of each 15-min data block (60 s window) were calculated (8 pairs of frequency and magnitude per participant).

2.3.3. *Surface electromyography (sEMG) for measuring spine muscle activity*

All participants were instrumented with six surface electromyography (EMG) surface electrodes to monitor the muscle activity of three back muscles bilaterally: thoracic erector spinae (RTS, LTS) 5 cm lateral to level T9, lumbar erector spinae (RLS, LLS) 3 cm lateral to level L1, and lumbar multifidus (RML, LML) 4 cm lateral to level L4 at a 45° angle. Before applying the electrodes, the skin was lightly shaved, abraded with tissue, and cleaned with a diluted isopropyl alcohol cleansing solution. For each muscle, two disposable electrodes (Ag–AgCl, Blue Sensor, Medicotest Inc., Ølstykke, Denmark) were placed over the muscle belly with a centre-to-centre inter-electrode distance of 2 cm. The raw EMG signals were differentially amplified, bandpass filtered from 10–1000 Hz and then digitally sampled at 1500 Hz using a 16-bit A/D converter with a resolution of ± 2V (Desktop DTS, Noraxon, Phoenix, AZ, USA; CMRR > 100 dB, input impedance > 100 MΩ).

EMG data were then processed by a custom program (Matlab version 2017b, The Mathworks Inc., Natick, Massachusetts, USA) to remove DC bias, high-pass filtered (30Hz) to remove heart rate contamination (Drake and Callaghan, 2006), full wave rectified, and smoothed with a 2nd order dual-pass Butterworth filter (2.5Hz cut off frequency (Brereton and McGill, 1998)). Signals were then normalized by subtracting resting EMG levels and dividing by maximum voluntary contraction (% MVC). Average normalized EMG was calculated over eight 15-min blocks throughout the trial.

2.3.4. *Ratings of Perceived Pain*

Ratings of Perceived Pain (RPP) were measured using a digital 100 mm visual analog scale using a custom desktop program (Matlab version 2017b The MathWorks, Natick, MA, USA). Participants were asked to rate their pain of their lower back by sliding a bar along a 100 mm continuous line with the following anchors: 0 mm = “no pain” and 100 mm = “worst pain imaginable”. When saved, the rating bars reset to zero so that past scores would not influence subsequent scores. The baseline rating was subtracted from each subsequent rating and the peak change in perceived pain was calculated.

2.3.5. *Seat pressure*

A pressure sensor array mat (LX210:40.40.02 Sensor, XSensor Technology Corporation, Calgary, AB, Canada) was fixed to the seat pan of the test chair during each session using Velcro™ tape. The origin of the sensor surface was always placed at the back right of the seat pan. Seat pressure data were sampled at a rate of 30 frames per second and were synchronized to the rest of the signals by external trigger. Data were processed with the X3 Pro Version 8.0 software (XSensor Technology Corporation, Calgary, AB, Canada) to calculate the peak pressure (N/cm^2), average pressure (N/cm^2), and contact area (cm^2) for eight 15-min blocks from the seat pressure distributions on each chair.

2.3.6. *Questionnaires*

Participants completed an Exit questionnaire after each experimental condition to rate their agreement to several statements related to their experience on a 5-point Likert scale (0 = strongly disagree; 5 = strongly agree). The questions on this questionnaire are included in Fig. 6.

2.4. *Data collection procedure*

Two experimental sessions were scheduled for each participant: one using the dynamic chair (dynamic condition) and one using a standard office chair (control condition). These sessions were scheduled at the same time of day to control for diurnal variation, and at least one day apart to prevent any carry over effects. The participants were block randomized to start with either the intervention or control chair using a random number generator in Excel (Version 14.4, Microsoft Office, Redmond, WA, USA). During the first session subjects were asked to

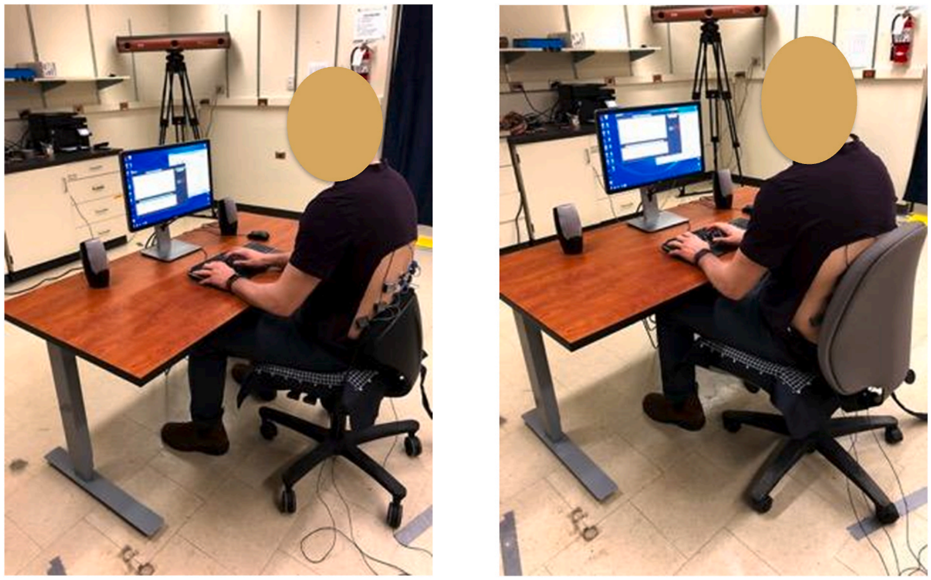

Fig. 2. Set up at the workstation in the dynamic (left) and control chair condition (right) respectively according to ergonomic guidelines with the pressure mat placed on the seat pan and the EMG and accelerometer sensors attached to the participant's back.

confirm study eligibility and complete informed consent. Following this, or upon arrival at the second session, participants were seated at the experimental workstation and the desk height, chair height, monitor height/depth and keyboard/mouse placement were adjusted according to ergonomic guidelines (Canadian Standards Association, 2000; Fig. 2). A baseline RPP was completed. Next, the participants were instrumented with EMG, and one 5-s resting trial (laying prone) and three 10-s isometric maximum voluntary contraction (MVC) back extension trials were collected. These MVC trials involved the participant being secured at the hips and legs on a therapy plinth extending their torso against the manually applied resistance of a researcher. Next, participants were instrumented with accelerometers. Four 5-s posture trials were completed to normalize spine angle data: upright standing, maximum flexion, maximum extension, and seated maximum flexion. Then, participants were seated, instrumented for the calf circumference measure, and a baseline measure was taken. In the following, participants watched a short video introducing the chair to be used in the session to provide standardized information with minimized bias to all participants clearly explaining the features and normal use of both chairs involved in this study. Each video was approximately 30 s long and was created by MB. The purpose of including two videos was to treat each condition as similarly as possible, and to avoid the perception of bias towards one chair over the other. The 2-h sitting trial began immediately after the video was shown during which participants completed a standardized typing task. RPPs were completed every 7.5 min throughout the duration of the trial starting at 0 and finishing at 120 min. Immediately upon completion of the sitting trial, the calf circumference measure was repeated. Finally, participants completed the exit-questionnaire and were de-instrumented.

2.5. Statistics

Descriptive statistics were presented as means and standard deviations and were calculated for all variables. SPSS statistical software (Version 22.0, IBM Corporation, Armonk, NY, USA) was used to conduct a two-way general linear model MANOVA between time (between eight 15-min blocks making up the 2 h sitting trial) and condition (dynamic vs control condition) for the dependent variables of normalized lumbar spine angle, pelvic angle, sagittal chair angle, standard deviation of the sagittal chair angle, range of the sagittal chair angle, coronal chair angle, standard deviation of the coronal chair angle, range of the coronal chair angle, number of fidgets, amplitude of fidgets, EMG of all 6 back muscles, average seat pressure, peak seat pressure, and contact area. If no significant interactions or main effects for a factor were found, the factor was removed from the model and a one-way ANOVA was completed on the remaining factor to increase power. For change in calf-circumference and peak change of RPP, a one-way MANOVA between chair conditions was completed to determine significance between chair conditions. A Wilcoxon signed-rank test was used to analyze the exit questionnaire responses. Statistical significance was set at $p = 0.05$ and to determine effect sizes partial eta squared (η^2) was calculated where 0.01 is considered small, 0.06 medium and 0.14 considered large effects (Cohen and Cohen, 1988). Interclass correlations ($ICC_{2,1}$) were calculated to assess within day and between day intra-rater reliability for the calf circumference measure where less than 0.50 is considered poor, between 0.5 and 0.75 moderate, between 0.75 and 0.90 good and above 0.90 excellent (Koo and Li, 2016).

3. Results

No significant interactions between time and chair condition or main effect of time were found for any time-varying outcome variables (Supplementary Table). Thus, time was removed from the model for each variable and a one-way ANOVA was completed to increase power. The data presented in the results section are the results from these one-way ANOVAs comparing between chair conditions.

3.1. Participant characteristics

Thirty-one participants were recruited. One participant was not able to return for the second session due to repeated scheduling conflicts and was excluded from the final analysis. Participant characteristics are presented in Table 1.

Table 1
Characteristics of participants ($n = 30$) that completed both sessions of the study.

	Mean (SD)	Range
Age (years)	24.2 (6.5)	19–56
Height (cm)	180.3 (6.2)	167.6–195.6
Mass (kg)	80.4 (14.3)	54.4–114.0
BMI (kg/m^2)	24.6 (3.7)	16.8–35.9

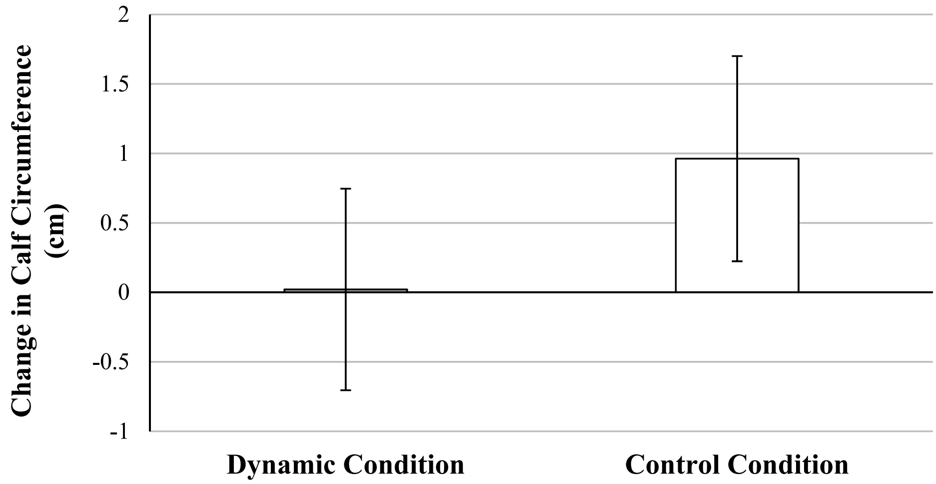

Fig. 3. Average change in calf circumference (cm) after the 2-h typing trial in both the dynamic and control condition ($p < 0.001$, $\eta^2 = 0.304$).

3.2. Change in calf circumference

A significant difference between chair conditions was found for change in calf circumference $p < 0.001$, $\eta^2 = 0.304$, Fig. 3). Calf circumference increased in both conditions; however, this increase was lower in response to the dynamic condition (0.02 ± 0.73 cm) compared to the control condition (0.96 ± 0.74 cm). The within day and between day intra-rater reliability of the calf circumference measures were found to be excellent (within day $ICC_{(2,1)} = 0.980$ 95% CI [0.895–0.999], between-day $ICC_{(2,1)} = 0.968$ 95% CI [0.700–0.998]).

3.3. Spine angles (tri-axial accelerometers)

There were no significant differences in the average spine angles between the dynamic (62.09 ± 19.70%ROM) and control condition (69.44 ± 11.87%ROM; $p = 0.103$, $\eta^2 = 0.048$; Fig. 4). There were also no differences in the pelvic angle between chair conditions. Participants sat with a −2.58 ± 10.72° pelvic angle in the dynamic chair and −2.02 ± 9.43° in the control chair ($p = 0.836$, $\eta^2 = 0.001$).

3.4. Spine movement (fidgets)

There were no significant differences in the average number of fidgets ($p = 0.396$, $\eta^2 = 0.034$) or the average magnitude of the fidgets ($p = 0.844$, $\eta^2 = 0.006$) between chair conditions. The average number of fidgets per minute in the dynamic chair was 9.92 ± 3.07 and 9.60 ± 2.60 in the control chair. The average magnitude of the fidgets in the dynamic chair was 2.61 ± 1.77° and 2.41 ± 1.51° in the control chair.

3.5. Seat pan orientation and movement

A significant difference between chair condition was found for the average angle of the seat pan tilt in the sagittal (forward-backwards) plane ($p < 0.001$, $\eta^2 = 0.485$, Fig. 5). Specifically, the seat pan was tilted forward when participants were seated in the dynamic compared to the control condition. There was also a significant difference for range of seat pan angle in the sagittal plane in the dynamic compared to the control condition (Fig. 5; $p = 0.003$, $\eta^2 = 0.140$). Similarly, there was a significant difference between conditions for the average standard deviation of seat pan angle in the sagittal plane ($p < 0.001$, $\eta^2 = 0.325$).

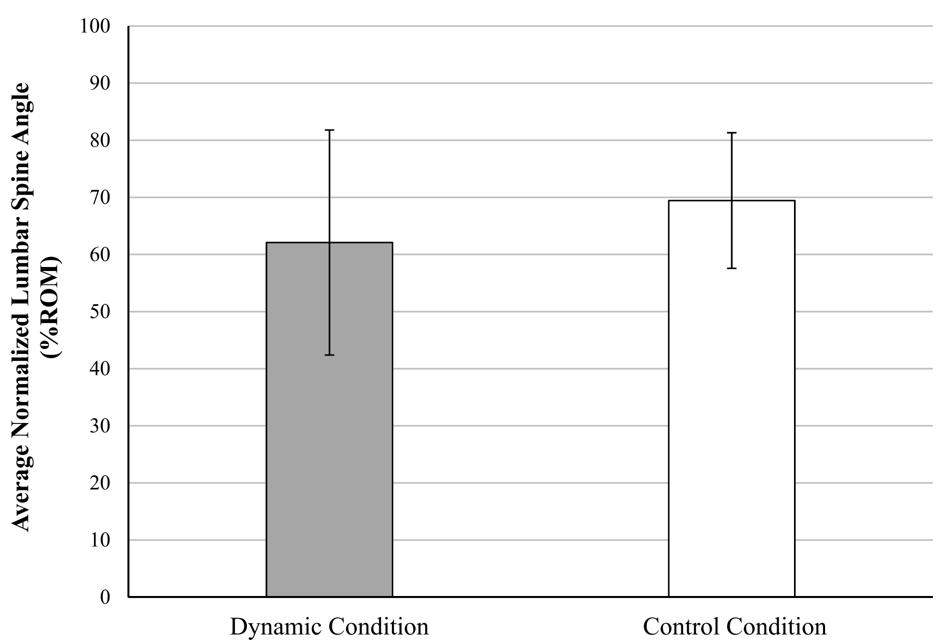

Fig. 4. Average normalized lumbar spine angle (%ROM) over the 2-h typing trial for both the dynamic and control conditions ($p = 0.103$, $\eta^2 = 0.048$).

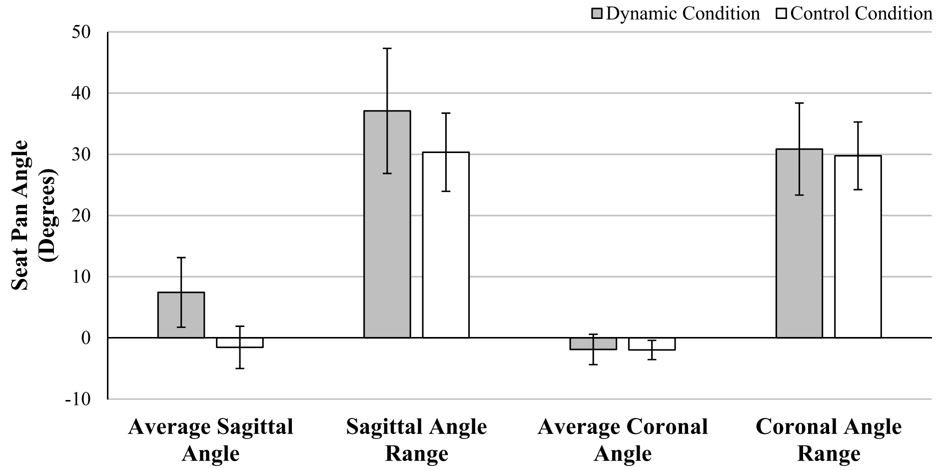

Fig. 5. Average seat pan angles and range with their standard deviations (error bars) in seat pan angle in both the sagittal (forward/backward) and coronal (side/side) plane of the chair over the 2-h typing trial in both the dynamic and control conditions.

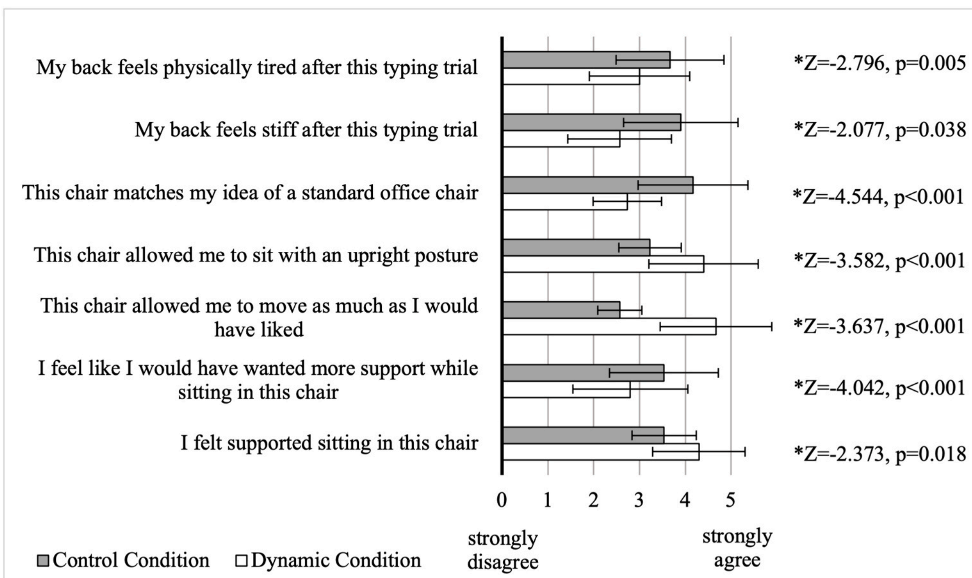

Fig. 6. Average values and standard deviations (error bars) for the Exit questionnaire responses on a 5-point Likert scale administered after the 2-h typing trial for thirty participants in both the dynamic and the control conditions.

This was larger in the dynamic chair (1.45 ± 0.89°) compared to the control chair (0.54 ± 0.30°).

In the coronal plane, a significant difference was found for average standard deviation in the dynamic (0.77 ± 0.34°) compared to the control condition (0.37 ± 0.97°; $p < 0.001$, $\eta^2 = 0.395$). No differences in the average angle of the seat pan and range were found between the dynamic (−1.89° ±2.48, 30.86° ±7.52) and control condition (−1.97° ±1.57; $p = 0.875$, $\eta^2 < 0.001$; and 29.76 ± 5.23°; $p = 0.520$, $\eta^2 = 0.007$).

Table 2
Average muscle activity (%MVC) and standard deviations of six low back muscles throughout the 2-h typing trial compared between the dynamic and control condition with the p-value and partial eta square presented.

Muscle	Average Muscle Activity %MVC (SD)		p-value	η^2
	Dynamic Condition	Control Condition		
RTS	2.94 (1.84)	3.74 (2.57)	0.173	0.032
RLS	2.39 (1.94)	3.47 (3.14)	0.115	0.042
RML	1.80 (1.70)	2.08 (1.80)	0.547	0.006
LTS	2.68 (2.09)	3.41 (2.75)	0.248	0.023
LLS	2.53 (1.99)	3.44 (3.32)	0.201	0.028
LML	1.67 (1.01)	2.25 (1.62)	0.101	0.046

3.6. *Average muscle activity during the prolonged sitting trials*

The muscle activity of all channels tested were very low for both chair conditions (1.80–3.74 %MVC) with no significant differences between chairs (p = 0.101–0.547; Table 2).

3.7. *Perceived pain ratings*

The peak change of PPR from baseline (0 min) in the low back region was found to be significantly different between the dynamic chair (9.95 ± 10.28 mm) and the control chair (18.33 ± 18.10 mm; $p = 0.031$, $\eta^2 = 0.077$). Perceived pain ratings continuously developed with time in both chair conditions, however, the magnitude of peak pain rating was much higher in the control chair.

Table 3
Average values, their p-value, and partial eta square for average pressure, peak pressure, and contact area between the subject and chair for all during the 2-h typing trial in both the dynamic and control conditions.

	Dynamic Condition	Control Condition	p-value	η^2
Average Pressure N/cm^2 (SD)	0.50 (0.07)	0.61 (0.11)	<0.001	0.287
Peak Pressure N/cm^2 (SD)	1.71 (0.61)	1.77 (0.42)	0.702	0.003
Contact Area cm^2 (SD)	1470.14 (199.34)	1325.06 (164.83)	0.003	0.140

3.8. Seat pressure

Average pressure and contact area were significantly different between the dynamic and control conditions (Table 3). There were no significant differences in peak pressure between the dynamic compared and the control condition (Table 3).

3.9. Exit questionnaire

The Wilcoxon signed-rank test showed that responses to all seven questions were significantly different between conditions (Fig. 6). When participants sat in the dynamic condition, they felt more supported, wanted less additional support, felt that the chair allowed them to move more and sit more upright, and felt less stiff and physically tired after the typing trial compared to the control condition, while the dynamic chair matched the idea of a standard office chair less than then control chair (Fig. 6).

4. Discussion

This study investigated the posture, movement and lower limb swelling of occupants sitting on a dynamic seat pan compared to a standard design. We found the dynamic condition mitigated calf swelling and increased seat-pan movement throughout a 2-h standardized typing task in comparison with the control condition. Spine angles, back muscle activity, and peak seat pressure was similar between conditions while average pressure was lower and contact area higher in the dynamic condition. PPR and subjective perceptions of the sitting experience were found to be significantly different between chair conditions.

4.1. Calf circumference

Increased calf circumference resulting from leg swelling secondary to venous pooling after periods of prolonged sitting has been previously highlighted in the literature (Chester et al., 2002; Seo et al., 1996). Additionally, Vena et al. (2016) showed that venous pooling rapidly occurs within the first 45 min of sitting. In our study we found that calf circumference remained largely unchanged over the 2-h sitting trial in the dynamic condition but increased by approximately 1 cm in the control condition (Vena et al., 2016). These results are consistent with a study conducted by Triglav et al. (2019) who used the same dynamic chair design and similar control (Triglav et al., 2019). There are two potential aspects of the seat pan design that could be leading to these findings. First, we hypothesized that by facilitating movement of the occupant, the lower limbs move more, promoting blood flow and thereby reducing pooling in the extremities. Our study did show increased seat pan movement in the dynamic compared to the control condition. While lower limb kinematics and/or muscle activity were not recorded, this could be indirect evidence that the lower limbs were moving driving this seat pan movement. Secondly, it could be that the movement capability of the seat pan leads occupants to adopt a more open hip angle even in static sitting, by choosing a more forward tilting orientation. This would reduce mechanical compression at the anterior hip leading to lower impedance of venous return from the lower limbs at the hip (Morishima et al., 2017). Hip angle was not measured in our study; however, we can assume there was less flexion at the hip given the more anteriorly rotated seat pan in the dynamic condition. We expect that the combination of a more open hip angle and increased chair movement as facilitated by the seat pan pivot mechanism may be enough to mitigate lower limb swelling in short durations of sitting and should be explored in larger field studies.

4.2. Spine posture and movement

The results of this study found that participants sat with similar spine posture in the dynamic and the control condition. This is comparable to results found from the investigations of van Dieen et al. (2001) who tested 2 variants of dynamic chairs that allowed movement in the sagittal plane only, finding no significant difference in spinal flexion (van Dieen et al., 2001). Similarly, Ellegast et al. (2012) concluded there were no significant reduction in lumbar flexion angles in four other novel variants of dynamic chairs, instead concluding that the task being performed significantly impacted posture and kinematics (Ellegast et al., 2012). However, it is important to notice that the standard deviation of the lumbar angle was larger in the dynamic than the control condition. The larger standard deviation suggests that the dynamic design feature that allows for seat pan tilt about a pivot point made it easier for participants to sit with a posture of their choice resulting in greater variation in the seat pan angle. Facilitating a greater range of postures, and therefore joint motion, during sitting may play a role in reducing potential negative effects of static loading on body.

It was hypothesized that participants would vary their posture more in the dynamic condition. While larger variation was noted in the dynamic condition, no significant differences in average spine posture were identified. This is consistent with previous studies of sitting that have found spine posture remains fairly static during prolonged exposures in laboratory-controlled studies (Gregory et al., 2006; Dunk and Callaghan, 2005, 2010; Beach et al., 2005). Similarly, there were no significant differences in the spine angle fidgets (number of times the spine flexion angle moved quickly away and back) between chair conditions. However, we only looked at fidgets in the sagittal plane. The significantly greater standard deviation of the seat pan angle in both sagittal and coronal planes suggest that there may have been more trunk movement occurring that was not captured.

4.3. Seat pan movement

The results from the accelerometer mounted on the seat-pan of the chairs in this study found the seat pan was tilted more anteriorly and moved more (larger standard deviation) in the dynamic condition relative to the control condition. Previous literature on anteriorly rotated seat pans have shown this feature to be associated with a decrease in LBP which is thought to be due to the promotion of increased lumbar lordosis (Gale et al., 1989; Gadge and Innes, 2007). At the same time, there was a significant difference in the standard deviation of the seat pan angle, indicating that participants were in fact actively moving in this plane, albeit continuously returning to a neutral position. This suggests that the design feature was used as intended – to increase movement. However, it is important to recognize that seat-pan movement and spinal flexion changes are not exclusively dependent on each other and do not always occur at the same time.

4.4. EMG

The results from the EMG placed on the lower back support the theory of minimal torso movement. Prolonged sitting in the dynamic condition during the standardized typing task did not result in any statistically significant differences in back muscle activity when compared to the control condition. The average EMG levels for all thirty participants in all six muscles in both chair conditions were very low, with magnitudes equal to or lower than 3% MVC. These low levels of muscle activity are consistent with seated torso EMG levels previously published on office chair seat pans in the literature (Gregory et al., 2006; O'Sullivan et al., 2013). This suggests that the demands of sitting on the dynamic seat pan feature (dynamic condition) are comparable to those in traditional office chair designs. These results are also similar to other dynamic chairs that have been tested where no differences in torso EMG level were detected between chair conditions (O'Sullivan et al., 2013). It appears that instead different tasks (i.e., reading, data entry, mousing etc.) drive differences in muscle activity (van Dieen et al., 2001; Ellegast et al., 2012); therefore the results of this study may only be generalizable to exposures primarily consisting of typing/data entry.

4.5. Pain

Steadily increasing perceived low back pain, consistent with many prolonged sitting studies (Greene et al., 2019; De Carvalho and Callaghan, 2011; Nairn et al., 2013; Schinkel-Ivy et al., 2013), was seen for both chair conditions in this study. However, participants reached significantly higher peak pain ratings in the control compared to the dynamic condition. These results suggest participants had a less painful experience seated in the dynamic condition while completing the typing task compared to the control condition. Similarly, O'Keefe et al., 2013 showed that pain decreased significantly more in the same dynamic condition than in a control condition in patients who already suffered from back pain related to prolonged sitting (O'Keeffe et al., 2013). Transient sitting-induced pain often resolves quickly once people can get up and move around from their chair. We propose that the reduction in pain observed in this study is due to the in-chair postural movement at the spine, pelvis and hips facilitated by the multi-axis tilt feature of the dynamic seat design during the sitting exposure.

4.6. Seat pressure

There was no difference in peak pressures between the chair conditions but significant differences in average pressure and contact area. The significantly lower average pressures and greater contact area in the dynamic compared to the control condition are likely directly related to the seat pan design. Specifically, the contoured seat pan design appears to provide a larger contact area and thus a better distribution of weight. Since pressure is the result of force divided by area, this is likely the factor driving the lower average pressure finding in the dynamic chair compared to the control chair. Further, study participants, on average, sat with the seat pan rotated forward by 8° which would transfer the ground reaction force of the upper body from the buttock to the feet and thus, reducing pressure at the buttock. At the same time, dynamic movement could have increased seat pressure momentarily resulting in an increased peak pressure. This could explain why the peak pressure was similar in both chair conditions.

4.7. Exit questionnaires

The exit questionnaire suggests that participants felt supported in both chairs although they felt significantly more supported in the dynamic condition. Similarly, they felt stronger about wanting more support in the control condition. This might be related to the participant's perceiving their back to be significantly stiffer and tired after the sitting trial in the control condition which might be related to a more constrained sitting posture. Participants indicated that they strongly agree that the dynamic chair allows them to move as much as they would have liked while rating this statement significantly lower for the control condition. They also felt that the dynamic condition allowed them to sit more upright which might be related to the angled seat pan position of the dynamic chair. However, participants were on average neutral about how the dynamic chair fits the idea of a standard office chair.

4.8. Strengths and limitations

There are several strengths to the design of this study. Chair conditions, despite obvious physical differences, were treated as similarly as possible to minimize bias. This included removing all identifying product logos, removing arm rests from the standard chair and creating an educational video for each chair highlighting an equal number of features and how they work. Chair conditions were conducted on separate days to ensure no carry over effects and the order of presentation was block randomized. Finally, the within day and between day test-retest reliability was calculated for the primary outcome measures, ensuring there was excellent reliability which improves the confidence of our results.

Methodological decisions were made, however, that do limit the generalizability of our findings. Only male participants were included in the study to reduce heterogeneity and increase the statistical power of our sample. This decision means that the results may not be applicable to females. There is evidence to suggest this is not the case for our primary outcome measure given that a similar study, with a population primarily of females, shows consistent calf circumference results (Triglav et al., 2019). Finally, while it was our intention to include a wide range of ages and body types, our recruitment resulted in a sample of younger (university aged) individual of normal BMI. Similarly, we controlled for the confounding effects of comorbidities by only including healthy individuals. This means that our results are not generalizable to older adults with clinical conditions. Finally, the sitting exposure tested in this study was only 2-h. Individuals sit for a much longer period of time than this in a typical workday. However, considering that the average worker likely stands up for some sort of break (coffee, lunch, bathroom etc.) around the 2-h time point, the duration tested in this study would at least be generalizable to one of these blocks of sitting. This limitation could be overcome in the future by using a field study design where the entire workday is studied.

5. Conclusion

The dynamic seat pan design mitigated calf-circumference increases and perceived low back pain during a 2-h exposure to sitting. This might be explained by the movement facilitating nature of the multi-axial seat pan which appears to have allowed participants adopt a more open hip angle and to move in all directions while keeping their overall spine posture and back muscle activity stable. Future studies should include the analysis of leg kinematics and muscle activity and aim to investigate a longer period of sitting, ideally in a field setting, with a more generalizable population.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.apergo.2021.103546.

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